Improving the Workflow of Collecting Shot Statistics for the Pomona-Pitzer Women's Water Polo Team

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Introduction

Background & Related Work

The Data

This project will use game film from the Pomona-Pitzer Women's Water Polo (PPWWP) 2026 season. Limiting the film to one season keeps the dataset manageable, while reducing variation in the roster and coaching staff, as players graduate, join, or leave the program. While Pomona-Pitzer is Division III, the PPWWP team plays Division I, II, and III teams throughout their season. Opponents in each division vary in difficulty, and thus I will only use the teams ranked in the final 2026 Collegiate Water Polo Association (CWPA) top 25. The CWPA national varsity rankings are determined by a committee of coaches and athletic representatives who rank teams based primarily on game results. Although Claremont-Mudd-Scripps (CMS) is not ranked in the top 25, I chose to include them, as they are PP's rivals and PPWWP's biggest Division III opponent. During the 2026 season, PPWWP played 14 games against 12 teams ranked in the top 25, with two of those opponents played twice. Additionally, PPWWP played against CMS five times, thus this project includes 19 games. The PPWWP coaching staff is most concerned with the top teams, and limiting the sample in this way addresses their needs accordingly.

The raw data consist of Hudl Sportscode CSV export files from 19 PPWWP games. Across the 19 files, there are roughly 12,000 event-level observations, with approximately 500–750 tagged events per game. Each observation, or row of data, is a tagged interval or event on the game timeline. Each observation includes its start time, duration, event category, and instance number. Tags include shots, goals, possessions, exclusions, power-play situations,

turnovers, player participation, and other game events. Because multiple tags can overlap during the same play, the exported files require additional organization before they can be used for shot analysis. Additionally, the files vary across games in the titles of tagged intervals/events, such as “PP Shot” vs “D Shot” (Dark) or “W Shot” (White), and the coded columns (i.e. some games have more columns than others). Therefore, this project requires standardizing the raw files, identifying PP shots and their relevant game-situation context, and combining these with the shot characteristics that need to be observed from the corresponding game film. The result will be a standardized workflow that produces a consistent dataset where each row is one PP shot.

Methodology: Data Processing, Workflow Development

Shiny Application

Results

Discussion, Ethics, and Limitations

Conclusion

Bibliography/References

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